Nkosinathi Hlophe

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Professional Summary

Data Scientist & AI Solutions Leader with 5+ years of experience delivering analytics, automation, and AI solutions across banking, research, and education. Proven ability to translate complex data into actionable business insights, deploy machine learning solutions at scale, and lead responsible AI adoption aligned to governance and privacy standards. Experienced in driving process efficiency, analytics monetization, and enterprise AI transformation within financial services.

Experience

FNB Eswatini — **eZulwini, Eswatini**
Data Scientist / Solutionist — *May 2025 – Present*

- Led automation and analytic solutions using Python, SAS, SQL, and Power Automate to reduce manual reporting time by about 20 % for some key operational reports.
- Lead cross-functional collaboration with business units to translate analytical outputs into strategic decisions.
- Contributed to the development and rollout of the bank's Artificial Intelligence (AI) strategy, supporting responsible AI adoption
- Deployed localized agentic AI solutions to improve employee experience and save time.
- Supported enterprise analytics standards, data governance, and ethical AI practices.

Standard Bank Eswatini — **Mbabane, Eswatini**
Data Scientist — *Oct 2021 – May 2025*

- Built and deployed **machine learning models** for cross-sell optimization and customer attrition analysis using Python and scikit-learn.
- Automated compliance, reconciliation and operational workflows using Python and Power automate, reducing manual intervention and operational risk.
- Designed and delivered executive-level Power BI dashboards, improving KPI visibility and decision-making.
- Migrated robotics process automation (RPA) from Power Automate and Workfusion to Python, improving scalability and performance.
- Co-developed and implemented the Bank's AI adoption roadmap and governance framework.
- Supported analytics governance initiatives, data ethics compliance, and adherence to privacy standards across the analytics function.

Eswatini Economic Policy Analysis and Research Centre (ESEPARC) — **Mbabane, Eswatini**
Data-Analyst — *Aug 2021 – Oct 2021*

- I conducted business data analyses using Excel and developed a recommendation system to inform strategic decisions.
- I also prepared detailed reports to communicate key insights to stakeholders.

Academic & Teaching Experience (Selected)

- Part-Time Lecturer – Physics & Mathematics, **University of Swaziland, Institute of Development Management** and full-time at **Ngwane Teacher Training College**
- Delivered undergraduate and diploma-level instruction in Physics, Mathematics, and Applied Analytics
- Designed assessments, practicals, and academic materials.

Education

African Institute for Mathematical Sciences (AIMS) — **Kigali, Rwanda**
African Master's in Machine Intelligence (AMMI) — *Sept 2019 – Apr 2021*

Stellenbosch University — **Cape Town, South Africa**
Master of Science in Mathematical Sciences — *Aug 2017 – Jun 2018*

University of Swaziland
Bachelor of Science in Mathematics and Physics *Aug 2011 – May 2015*

Technical & Leadership Skills

Analytics Leadership:	Data Governance, Ethical AI, Model Lifecycle Management, Team Development
Programming:	Python, R, SQL, VBA, Git
Machine Learning:	Scikit-learn, TensorFlow, LSTM, Regression, Classification, Forecasting
Automation:	Power Automate, Selenium, Python RPA
Visualization:	Power BI, Matplotlib, Seaborn, Excel
Cloud & Big Data:	AWS (Certified), Azure (Working Knowledge), Data Monetization Pipelines

Selected Data Science Projects

- **Adversarial Texture Synthesis with Prototypes** - Built GAN-based models for pattern synthesis and representation learning.
- **Power Output Prediction for Wind Farms** - Developed LSTM and SVR forecasting models to predict wind speed and energy output, supporting operational planning.

Certifications

- AWS Certified Cloud Practitioner (2022)
- SQL for Data Science – Coursera (2023)
- Data Structures & Algorithms in Python – Udacity (2019)
- Introduction to Machine Learning – Coursera (2019)

Workshops Attended

Stellenbosch University **Cape Town**
Deep Learning Indaba *Sep 2018*

Presented a poster on masters project on power output prediction for wind-farms using machine learning

University of Witwatersrand **Johannesburg**
Mathematical in Industry Study Group *Jan 2018*

Developing Mathematical Models for solving real life problems. Presented projects on gas extraction from Lake Kivu and removing flaws in plate glass

Leadership & Awards

Eswatini Chess Federation
Vice President *Nov 2021 – Apr 2024*

Eswatini Chess Federation
Chess Development Instructor *Mar 2019 – Present*

References available on request